# Direct Proof

Key Topic

Some theorems can use a **direct proof** which does not require any 'tricks' (or logical equivalences).

Outline for a Direct Proof of $P \Longrightarrow Q$:

Suppose P

.

. Work

.

Therefore Q

# Direct Proof

Example

1) Prove the following statement: If $x \in \mathbb{Z}$ is odd, then $x^2$ is odd.

2) Let $a, b, c \in \mathbb{Z}$. If $a|b$ and $b|c$, then $a|c$.

TopHat: Sort the following lines into a correct proof.